Lemma von Bezout

$$ax + by = \gcd(a,b)$$

Proposition: Suppose $a, b \in \mathbb{N}$, then there exists $x, y \in \mathbb{Z}$ such that

$$ax + by = \gcd(a,b)$$

proof: Consider

$$S = \{ax+by \mid x,y \in \mathbb{Z},\ ax+by > 0\} \subseteq \mathbb{N}$$

Note $a, b \in S \Rightarrow S \neq \emptyset$

By Archimedian principle there $\exists$ a minimal Element of S. Set $d = \min(S)$.

Claim: $d = \gcd(a,b)$

Notice $d = ax_0 + by_0$ for $x_0, y_0 \in \mathbb{Z}$ (1)

Claim: $d \mid a$

If $d \nmid a$ then $a = dq + r$ with $0 < r < d$

$(1) \cdot q$: $dq = aqx_0 + bqy_0$

$\Rightarrow a - r = aqx_0 + bqy_0$

$\Rightarrow r = (1 - x_0 q)a + (-qy_0)b$

$\Rightarrow r \in S$ and $r < \min(S) = d$ $\lightning$

analog $d \mid b$.

So $d = ax_0 + by_0$ for $x_0, y_0 \in \mathbb{Z}$, $d|a$ and $d|b$.

Let $c \in \mathbb{N}$ such that $c|a$ and $c|b$

$\Rightarrow c | ax+by \quad \forall x,y \in \mathbb{Z}$

In particular $c | ax_0 + by_0 \Rightarrow c|d$

$\Rightarrow d = \gcd(a,b)$ (d must be the greatest common divisor, cause an arbitrary divisor c of a and b divides d) $\square$